UTM UNIVERSITI TEKNOLOGI MALAYSIA | Faculty of Computing

SEMESTER II 2025/2026 SECD2613-04

SYSTEM ANALYSIS AND DESIGN

LECTURER'S NAME

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GROUP PROJECT 1 - GROUP 2

GROUP PROJECT : UTM TRANSPORT APP

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1.0 Introduction

Universiti Teknologi Malaysia (UTM) is based in Malaysia, which is located in a tropical climate with extreme hot temperatures and constant rains (Verkaik et al., 2018). As one of the best technological universities in the world, there are many students who move around every day from their residences to the various faculties. Therefore, an efficient campus transportation is no longer just a convenience but a vital necessity. However, the manual schedule and overcrowding during peak hours of the bus leads to significant inefficiencies in student mobility.

In response to the global shift towards 'Smart Campus' initiatives (Puppala, 2024), our efforts focus on bringing the physical and virtual sides of campus life together through the use of real-time data analysis and mobile applications to maximize efficiency and engagement within the campus ecosystem. The UTM Transport App is designed to revolutionize campus transport systems through a mobile application that will facilitate real-time bus tracking and allocation of seats.

Besides, the UTM Transport App integrates a dedicated student carpool module that optimizes campus logistics through secure data fusion. The app ensures safety by strictly limiting access to verified student identities via student's UTM email. In addition, the implementation of an intelligent route planning system will enable the students to make immediate and well-informed choices concerning the routes they take each day. This project is consistent with the university's goal of implementing cutting-edge technology solutions for improving student welfare and sustainability.

In the end, the aim of UTM Transport App is to make the campus more accessible to everyone, and to enable UTM students to move around the campus with ease. By combining all these important services within one platform, this app will pave the way towards other smart infrastructures of the faculty.

2.0 Background Study

2.1 Organization Background

Universiti Teknologi Malaysia (UTM) is a premier public research-intensive university in Malaysia and is responsible for handling the daily transport requirement of more than 25000 students throughout its huge Skudai campus. The university operates an internal shuttle bus service running different routes within the university campus connecting various residential colleges, faculties, and administration blocks.

As a technological university, UTM has recently been prioritizing "Smart Mobility" as one of the pillars in the efforts towards sustainability. With the adoption of "Smart Mobility", the university seeks to utilize digital innovation in promoting students' wellbeing. The suggested UTM Transport App will serve as the tool necessary in facilitating this process.

2.2 Case Study

Nowadays, UTM transport management mainly depends on fixed timetables and coordination. However, due to the high student density, this approach does not allow for dynamism in meeting the student's demands, making it hard for the students to know when the buses are arriving. As a result, students have been forced to wait in long lines at bus stops in the hope that they will get seats in time.

Unfortunately, this problem is further amplified by the challenging tropical climate in Malaysia. The Department of Environment (2023) confirms that the temperature in the Johor region, where the campus is located, often peaks at 34°C during the academic day and sometimes experiences heavy rains due to monsoons. Studies have shown that such extreme weather conditions affect academic performance and the wellbeing of the students (Nor Azli et al., 2023) .

Moreover, during the peak hours, the number of people waiting for the bus exceeds the number of available seats forcing the students to wait at the bus stop in vain, with no alternative like coordinated carpooling. Currently, students carpooling occurs randomly using messaging applications. In addition, there is no feedback system, meaning that the students do not get notifications whenever there are service failures like delayed or dirty buses due to lack of reporting mechanisms.

3.0 Problem Statement

At a campus as huge as Universiti Teknologi Malaysia (UTM), transportation is not just a "nice-to-have" feature, it is the heartbeat of our daily academic lives. However, the current transportation system tends to be fragmented and traditional that just cannot keep up with the intense schedule of a modern student. For most of us, arranging our traveling schedule is not the smooth transition it should be. It has become a daily source of stress and friction that drains our focus and productivity. To alleviate this issue, we need to address five specific "pain points" that are currently making life harder for the UTM community:

1. **The "Wait-and-See" Game (Bus Arrival Uncertainty)**

   Right now, the biggest headache is the total lack of transparency. We are forced to stand at bus stops for who-knows-how-long, often sweating under the scorching sun or getting soaked in a heavy rain, without any idea if the bus is two minutes away or twenty. This uncertainty causes a domino effect, such that we end up late for crucial lectures, miss the start of lab sessions, and live in a constant state of "transit anxiety" that makes it impossible to plan our study groups or even a simple lunch break effectively.

2. **The Morning Scramble (Peak-Hour Seat Shortages)**

   During the morning and evening rush, the campus transport system basically turns into "survival of the fittest" mode. This is because we have no way to check how full a bus is in advance, we often have to watch bus after bus drive past because they are packed with people from the previous stations. It is not just about the delay, but also about the discomfort of being squeezed into overcrowded vehicles and the safety risks that come with it, leaving those at later stops completely stranded.

3. **The "Lost" Seats (Inefficient Carpooling)**

   So many of us are willing to share rides or land a helping hand to ease the load on the buses, but there is not a safe, official way to make that happen. We are currently stuck using disorganized social media chats or word-of-mouth, which is awkward and unreliable. This "fragmented carpooling" means thousands of seats go empty every day while other students are left waiting at the station or walking with their bare feet, wasting a huge opportunity to build a better campus community and lower our carbon footprint.

4. Lost in the Maze (No Route Planning Tools)

UTM is huge, and for "freshies" or visitors, it can feel like a total maze. Without a smart tool to help plan a route, students struggle to find the fastest way to get from their residential colleges to distant faculties or admin buildings. We end up wasting time taking long, inefficient paths or walking much further than we need to, simply because we do not have a map that shows us how to navigate the transit transfers correctly.

5. The "Wall of Silence" (Lack of Feedback)

The most frustrating part of all this is that when things go wrong, we have no real voice. There is not a dedicated, systematic way to report things like a bus being consistently late, a vehicle being dirty, or safety concerns with driving. Without this feedback, the university admin is essentially "flying blind," making it impossible for them to see where the service is failing or how to fix it. This silence leads to a cycle of the same problems happening semester after semester with no end in sight.

4.0 Proposed Solutions

a) Technical Feasibility

In terms of hardware, we currently use laptops with 16GB RAM and powerful multi-core processors for development purposes. This is sufficient for developing the system, as these computers will easily manage the coding of the six main modules. For the user device, students at UTM will use the smartphone that is already equipped with GPS and internet services. There is no need for students to pay extra money to obtain these resources because they already have such a device available to them. However, the existing campus buses do not currently have hardware for providing location information. The technology required for Live Tracking will be achieved by equipping all buses with GPS transponders to provide their location to our system. At the moment, we do not own any physical servers for hosting the application. Rather than buying expensive equipment, we will use Cloud Infrastructure (AWS), which means that we can increase our storage space immediately upon more students using the service.

In terms of software, in order to make sure the application is accessible to all students, we will have to use software such as Flutter and React Native that allows us to develop one solution compatible with both Android and iOS mobile devices. However, we noticed that the software solutions that we currently possess will not be enough for dealing with real-time data management, and that is why we will have to use Firebase as our cloud backend solution in order to process the data in real time. Also, in order to integrate the Route Planning module, we will utilize Google Maps API.

b) Operational Feasibility

Firstly, a system should have enough human resources to operate. Fortunately, all bus drivers will have no need to be specially trained since the GPS trackers will work automatically when driving. Further, the continuous process of maintaining, updating the software, and monitoring the network can easily be done through the IT department of the campus itself, and could even involve talented students of computing faculties of the campus, who would not require any external contractors to do the job.

Secondly, a system can only be maintained when it is supported by the people. Long waits at bus stops due to the delayed schedule are an annoying problem that each student faces regularly. The transport system will solve a burning issue for many students as it will provide the exact information about the current position of each bus. Lastly, students already use their smartphones for everything in daily life. Therefore, they will find it easier to accept the new system rather than ignore it. The system will gain immediate popularity among users who will not oppose it.

c) Economical Feasibility

For the decision to consider the smart transport system to be financially beneficial to the university, the comparison should be made between the amount of money needed to set up the system and the predicted income over four years. The financial model breaks down the total expenses into two major components which is the one-time startup cost and annual operating costs. The startup involves installing the mobile app and purchasing GPS technology equipment for the buses, costing RM 33,550 in total. Besides that, after launching the mobile app, there will be an annual fee to support cloud servers and system maintenance equal to RM 8,250. But the project itself generates its revenue from advertising within the system and fees for using the carpool platform. Though the improvement of well-being for students cannot be evaluated financially, the income generated from the system is quite obvious and likely to grow exponentially every year due to higher user traffic.

Since money gradually loses value over time, the financial model uses a calculation to determine the present value of future income and expenses. Despite the fact that the model applies the calculations quite strictly, the resulting financial outlook is exceptionally promising. Our system charges a lower commission rate than Grab. We aim to earn RM18,000 from the carpool feature by estimating a daily revenue of RM50. Furthermore, we expect the app to bring in approximately RM1,400 in monthly advertising income.

The project is supposed to cover all of its own costs and start showing profits by the end of the second year with a total income of RM 8,283. By the end of the fourth year, the whole system will bring total profits amounting to RM 48,378. In order to reliably confirm the project's potential, we calculate the overall profitability by dividing the total profit by the start-up costs. As a result, the overall profitability indicator reaches 1.44. Being greater than one, this figure indicates that the project brings more profits than risks.

d) Cost - Benefit Analysis (CBA)

Estimated Costs	
Hardware (GPS Trackers)	5,500
Software Function Development	25,000
Cloud Hosting & Maps API	3,800
System Maintenance & Updates	2,500
Admin Support	1,200

Estimated Benefits	
Platform Revenue	18,000
Advertising Revenue	17,000

Assumptions	
Discount rate	10%
Sensitivity factor(cost)	1.1
Sensitivity factor(benefits)	0.9
Annual change in production costs	5%
Annual change in benefits	7%

COSTS		Year 0	Year 1	Year 2	Year 3	Year 4
Development Cost (One-time)						
Software Function Development	25,000	**27,500**				
Hardware (GPS Trackers)	5,500	**6,050**				
Total (Development Cost)		**33,550**				
Production Cost						
Cloud Hosting & Maps API	3,800		4,180	4,389	4,608	4,839
System Maintenance & Updates	2,500		2,750	2,888	3,032	3,183
Admin Support	1,200		1,320	1,386	1,455	1,528
Annual Production Costs			**8,250**	**8,663**	**9,096**	**9,550**
(PRESENT VALUE)			**7,500**	**7,159**	**6,834**	**6,523**
ACCUMULATED COSTS			**41,050**	**48,209**	**55,043**	**61,566**

BENEFITS		Year 0	Year 1	Year 2	Year 3	Year 4
Platform Revenue	18,000		16,200	17,334	18,547	19,846
Advertising Revenue	17,000		15,300	16,371	17,517	18,743
Annual Production Costs			**31,500**	**33,705**	**36,064**	**38,589**
ANNUAL BENEFITS (PRESENT VALUE)			**28,636**	**27,855**	**27,096**	**26,357**
ACCUMULATED BENEFITS			**28,636**	**56,492**	**83,587**	**109,944**
GAIN OR LOSS			**(12,414)**	**8,283**	**28,545**	**48,378**
PROFITABLE INDEX		**1.4420**				

5.0 Objective

The UTM Transport App is more than just a technical upgrade, it is a commitment to transforming the student experience from one of frustration to one of convenience. Our goal is to build an ecosystem that values a student's time as much as their education. To achieve this, the project is built around six human-centric objectives:

1. **Giving Students Their Time Back (Live Tracking)**

   The primary goal is to end the "guessing game" at the bus stop. We will implement a live tracking module that provides the exact real-time location of every campus bus. By offering precise Estimated Times of Arrival (ETA), we empower students to stay in the library or cafe until the very last moment, effectively eliminating the anxiety of waiting in the heat for a bus that may or may not be coming.

2. **Restoring Predictability to the Daily Commute (Seat Availability Monitor)**

   No student should have to wait anxiously at a stop, only to watch a completely full bus drive past when they have an exam in 20 minutes; therefore, we aim to develop a live Seat Availability Monitor that allows students to check exactly how much space is left on approaching buses in real-time, removing the guesswork and panic from the commute and transforming it into a managed, respectful transit experience where students can plan their journeys with confidence.

3. **Building a Community of Shared Rides (Carpool Module)**

   We want to move away from the "every person for themselves" mentality. By designing a secure, student-only carpool module, we will match student drivers with passengers heading the same way. This objective is about creating a flexible "safety net" for the campus transport network while helping us all work together to reduce our collective carbon footprint.

4. **Navigating with Confidence (Intelligent Route Planning)**

   Whether it is a freshman’s first week or a senior heading to a new faculty building, navigating UTM should be effortless. We will integrate a smart route planning tool that calculates the best path based on time, distance, and transfer options. The goal is to provide a "GPS for campus life" that removes the stress of getting lost and ensures students always find the most efficient way to their destination.

5. Creating a Culture of Accountability (Feedback & Rating)

For too long, student complaints have gone unheard. We will establish a transparent feedback and rating system where students can instantly score their ride and report issues like delays or vehicle conditions. This turns every student into an active participant in improving the system, ensuring that the transport administration remains accountable and responsive to our actual needs.

6. Simplicity Through Design (Figma Prototype)

Technology should solve problems, not create new ones. Our final objective is to design an intuitive mobile interface using Figma. We are focusing on a "frictionless" user experience, taking complex data like GPS coordinates and seat numbers and turning them into simple, actionable information that any student can understand at a single glance.

7. Fostering a Safe and Trusted Community (Verified User Authentication)

A smart campus must also be a safe campus. To protect our users from scams, especially within the carpool module, and to ensure the platform is used responsibly, we will implement a strict, student-only authentication system. By linking every account directly to a verified UTM identity, we create a closed, high-trust environment. This guarantees that every carpool match, seat booking, and feedback report is tied to an authenticated user, drastically reducing the risk of misuse and making it easy to track and resolve any incidents.

6.0 Scope of the Project

- **User Management Module**

  This module focuses on account management functions such as account creation, authentication, and user-profile maintenance to ensure that users of the application are indeed verified UTM students. They need to sign up using UTM email. So, users just need to register their account and click on the "Remember Me" checkbox if one wants quick future logins.

- **Live Tracking Module**

  This module offers users the ability to view live information on campus buses' routes and locations. The information displayed consists of the actual location of the bus on the map, along with an Estimated Time of Arrival at their current station. The module enables users to pick any bus on the map and observe its movement, along with its plate number.

- **Seat Availability Monitor Module**

  This module allows users to view real-time seat availability of approaching campus buses. The information displayed includes the current occupancy level (available, moderate, or full). So, users can select a bus and check its seat status along with its route, helping them decide whether to board or wait for the next bus.

- **Carpool Module**

  This module helps in making travelling on the campus flexible since it offers the students the ability to form rides with other students. Users have two options in this module either to form their own carpool ride or join others. Users can go through the list of available rides and find someone who matches and then contact the driver via his/her contact details.

- **Route Planning Module**

  This module acts as the smart guide in planning how users can travel on the campus in the best way. The fastest route options will be provided together with step by step instructions as well as any required bus transfer along the way. Users can simply input the destination point from the point of departure to access various route options.

- **Feedback & Rating Module**

  This module allows the users to help in improving campus transport through feedback. Here, users can rate their experience riding the bus and report problems such as lateness and overcrowding among others. Once the trip ends, a rating window pops out where users can rate the bus.

7.0 Project Planning

7.1 Human Resources

Name	Position	Responsibility
DR. NAZMONA BINTI MAT ALI	Advisor	Provides ongoing academic and professional feedback to the project for successful achievement of the desired objectives.
DARREN CHUA BOON YEE	Project Manager	Controls all aspects of the project management process, including planning and implementation.
TAN YU KAI	System Analyst	Collaborates with stakeholders in determining their needs and defining requirements for systems to achieve project objectives.
CHEW JIAN HUI	Front-End Web Developer	Designs and implements front-end applications, creating interactive and visually appealing web pages from design concepts and improving their performance.
PON XUEN LIN	Back-End Web Developer	Designs and implements back-end services, such as databases and APIs, as well as performs data processing and application logic implementation.
TAN IRENE	UI/UX Designer	Designs wireframes and prototypes and creates visual design components like layout, color, and typography to enhance user experience and maintain design uniformity.

ANG LE SHAN	Quality Assurance	Conducts testing and evaluation of the system to meet quality requirements by analyzing problems with functionality, usability, performance, and security.

7.2 Work Breakdown Structure (WBS)

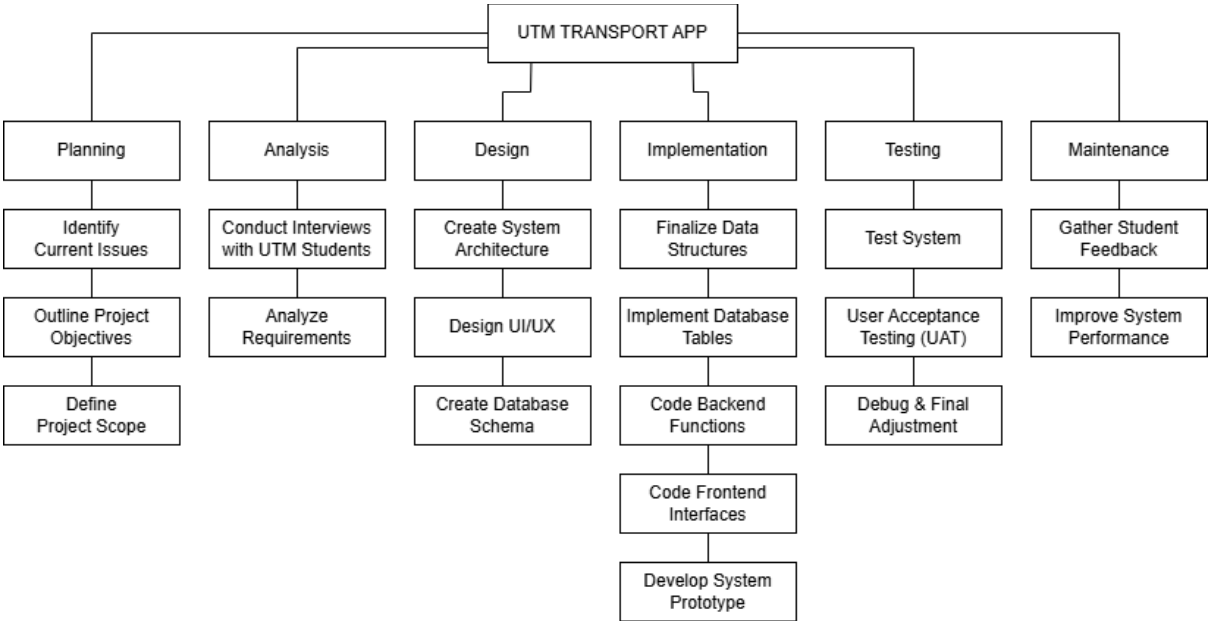

Figure 7.2.1 Work Breakdown Structure (WBS)

7.3 PERT Diagram

ID	ACTIVITY	PREDECESSOR	DURATION (DAYS)
A	Identify Current Issues	None	2
B	Outline Project Objectives	A	2
C	Define Project Scope	B	2
D	Conduct Interviews with UTM Students	C	5
E	Analyze Requirements	D	2
F	Create System Architecture	E	4
G	Design UI/UX	F	3
H	Create Database Schema	G	3
I	Finalize Data Structures	H	2
J	Implement Database Tables	I	3
K	Code Backend Interfaces	J	10
L	Code Frontend Functions	J	9
M	Develop System Prototype	K, L	4
N	Test System	M	3

O	User Acceptance Testing (UAT)	N	4
P	Debug & Final Adjustment	O	3
Q	Gather Student Feedback	P	2
R	Improve System Performance	Q	3

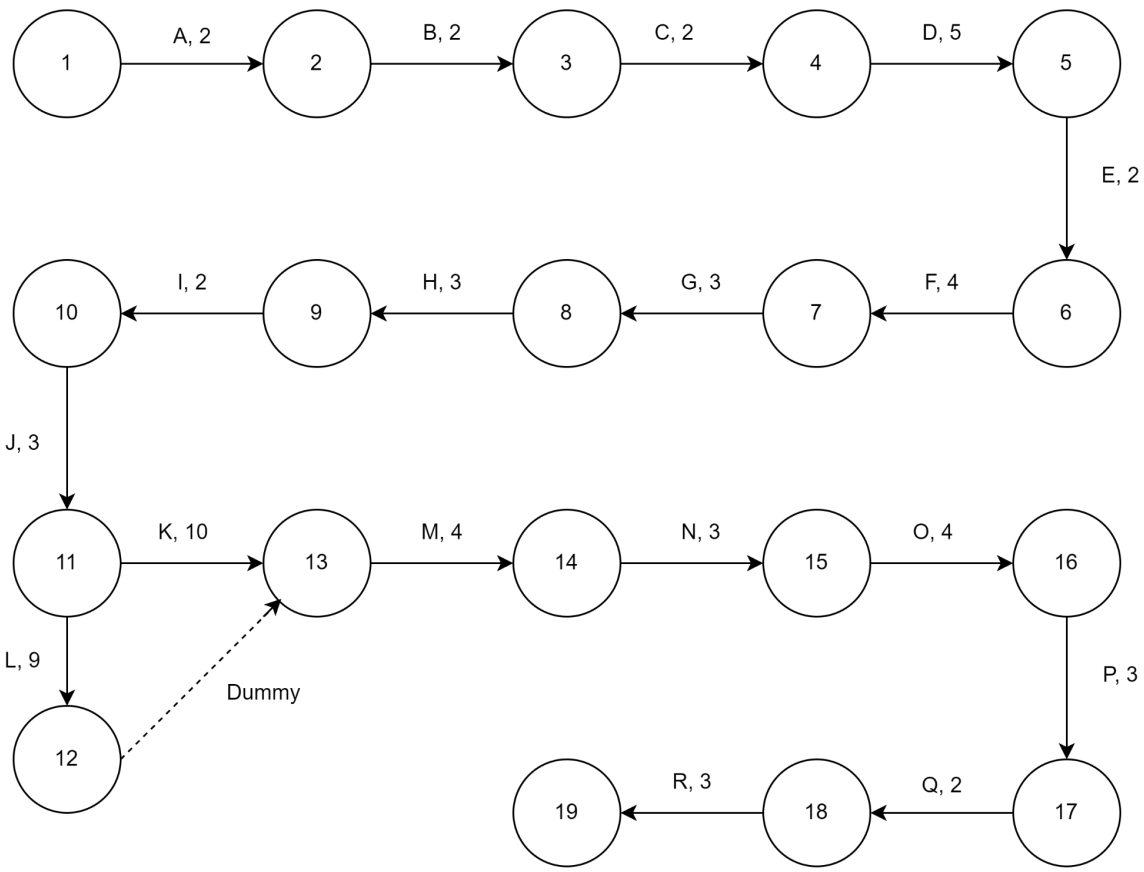

Figure 7.3.1 Pert Diagram

Note: All the durations are in days.

Path 1: A-B-C-D-E-F-G-H-I-J-K-M-N-O-P-Q-R

Length = 2+2+2+5+2+4+3+3+2+3+10+4+3+4+3+2+3

= 57 days

Path 2: A-B-C-D-E-F-G-H-I-J-L-M-N-O-P-Q-R

Length = 2+2+2+5+2+4+3+3+2+3+9+4+3+4+3+2+3

= 56 days

Since the Path 1 has a longer duration compared to the Path 2, thus the Path 1 is the critical path for this project. The shortest amount of time needed to complete this project is 57 days.

7.4 Gantt Chart

GANTT CHART

PROJECT TITLE	UTM Transport System Project Schedule		
PROJECT MANAGER	Darren Chua Boon Yee	DATE	6-Apr-2026

WBS NUMBER	TASK TITLE	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
1	**Planning**				
1.1	Identify Current Issues	6-Apr-2026	7-Apr-2026	2	100%
1.2	Outline Project Objectives	8-Apr-2026	9-Apr-2026	2	100%
1.3	Define Project Scope	10-Apr-2026	11-Apr-2026	2	50%
2	**Analysis**				
2.1	Conduct Interviews with UTM Students	12-Apr-2026	16-Apr-2026	5	0%
2.2	Analyze Requirements	17-Apr-2026	18-Apr-2026	2	0%
3	**Design**				
3.1	Create System Architecture	19-Apr-2026	22-Apr-2026	4	0%
3.2	Design UI/UX	23-Apr-2026	25-Apr-2026	3	0%
3.3	Create Database Schema	26-Apr-2026	28-Apr-2026	3	0%
4	**Implementation**				
4.1	Finalize Data Structures	29-Apr-2026	30-Apr-2026	2	0%
4.2	Implement Database Tables	1-May-2026	3-May-2026	3	0%
4.3	Code Backend Interfaces	4-May-2026	13-May-2026	10	0%
4.4	Code Frontend Functions	4-May-2026	12-May-2026	9	0%
4.5	Develop System Prototype	14-May-2026	17-May-2026	4	0%
5	**Testing**				
5.1	Test System	18-May-2026	20-May-2026	3	0%
5.2	User Acceptance Testing (UAT)	21-May-2026	24-May-2026	4	0%
5.2	Debug & Final Adjustment	25-May-2026	27-May-2026	3	0%
6	**Maintenance**				
6.1	Gather Student Feedback	28-May-2026	29-May-2026	2	0%
6.2	Improve System Performance	30-May-2026	1-Jun-2026	3	0%

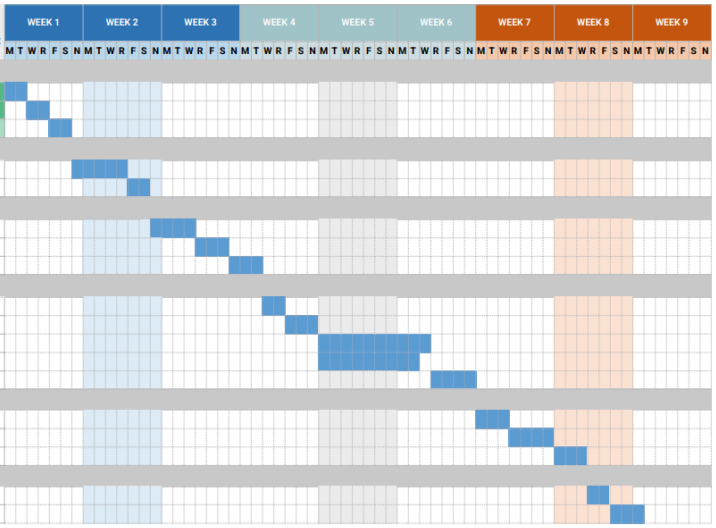

Figure 7.4.1 Gantt Chart

8.0 Benefit and Overall Summary of Proposed System

The UTM Transport App is not just about adding another icon to a student's phone, it is about fixing the broken parts regarding transportation of our daily lives on campus. By embracing a digital-first approach, this system offers real, tangible wins for both the students and the university administration.

8.1 Key Benefits of the System

The system provides several key benefits that improve campus transportation for UTM students by offering a centralized platform for registration, account management, and transport-related services. This makes the overall user experience more convenient and efficient.

One of the main features is real-time bus tracking, which allows students to view live bus locations and estimated arrival times. This helps users plan their journeys better and reduces waiting time. In addition, the carpool module gives students more flexibility by allowing them to create or join ride-sharing arrangements, making campus travel more cost-effective and convenient. There is also a feature providing users with suggestions concerning possible route planning with clear instructions for each step.

Furthermore, the feedback and rating module enables users to report issues and evaluate their travel experience, supporting continuous improvement of the system. The platform also ensures security by requiring users to register with verified UTM email accounts, ensuring that only legitimate students can access the services.

8.2 Overall Project Summary

Frankly speaking, the current transport situation at UTM is a genuine struggle that messes with our focus and our schedules every single day. The problem is not just that we need more buses, it is that we need better information and coordination.

The UTM Transport App is our plan to solve that. By taking those old, manual headaches and making them automatic, we are creating a much smoother flow for everyone on campus. Our research and feasibility studies confirm that this is not just a "nice-to-have" idea, it is technically doable, urgently needed, and a smart investment that will save the university time and money in the long run.

Ultimately, this project is about building a Smarter UTM with a strong transportation system. We are choosing to value our time, our safety, and our success by ensuring that the journey across campus is no longer a barrier to our education, but a seamless part of it.

9.0 Reference

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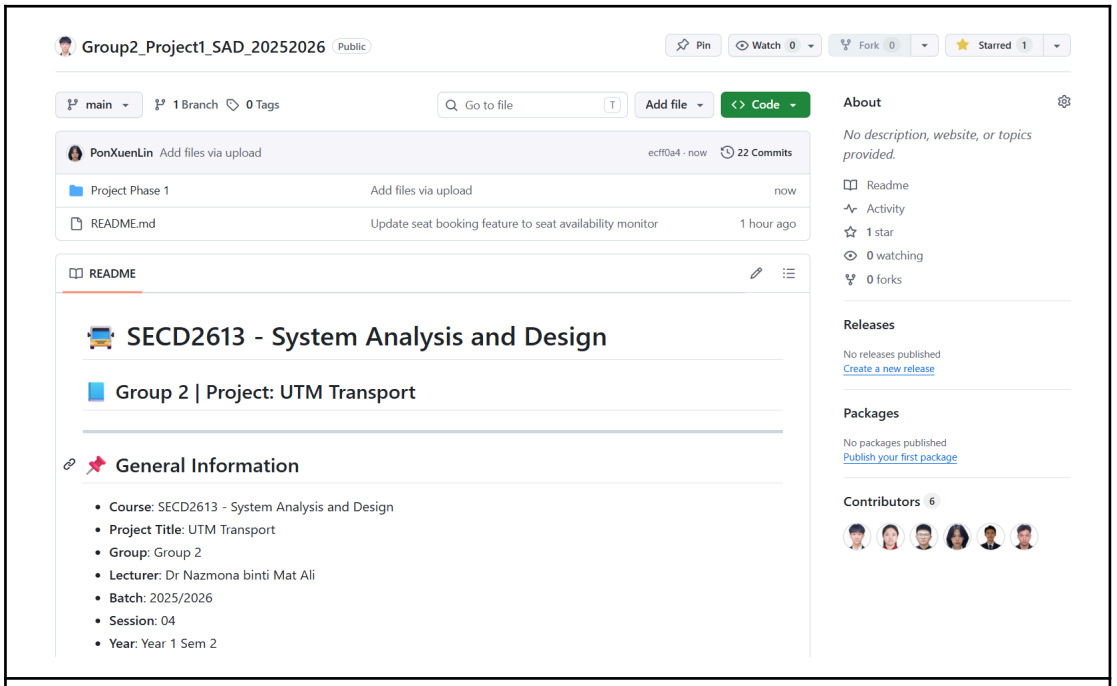
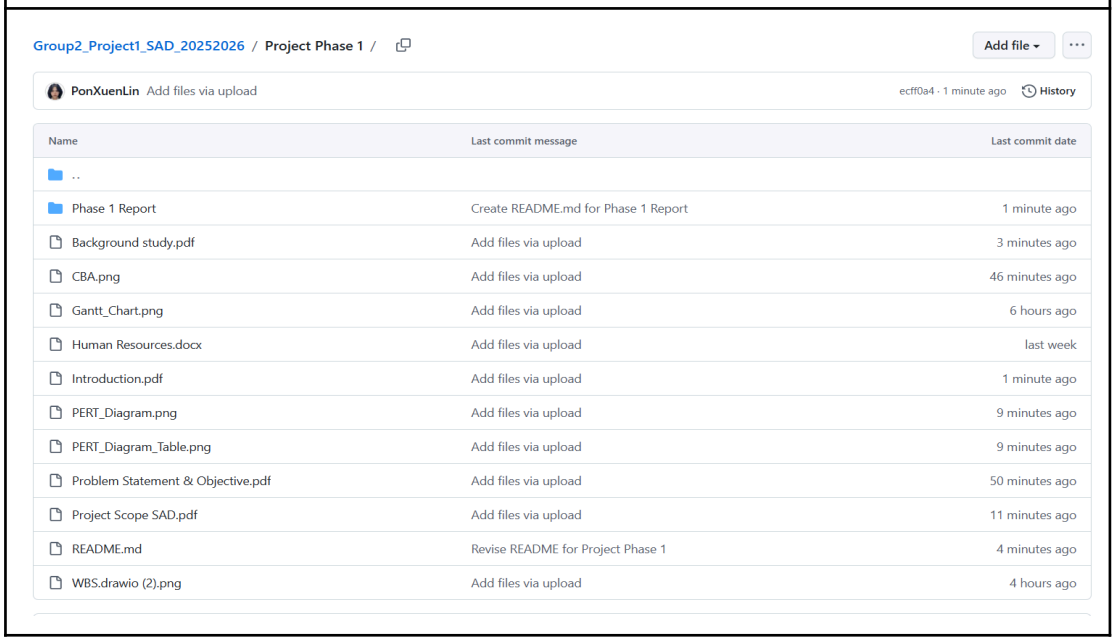
10.0 Appendix

- Github URL Link:
  https://github.com/Dc-by1/Group2_Project1_SAD_20252026
- Kanban Board Link:
  https://github.com/users/Dc-by1/projects/1
- Repository Snapshot: